

Create data base and tables

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'employees' selected. The main query editor contains the following SQL script:

```
1 • create database employess;
2 • use employess;
3 • create table employee(employee_id int,
4   employee_name varchar(50),
5   gender enum("M","F"),
6   age int,
7   hire_date date,
8   designation varchar(100),
9   department_id int,
10  location_id int,
11  salary decimal (10.2)
12
13
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	20:01:22	create database employess	1 row(s) affected	0.047 sec
2	20:01:22	use employess	0 row(s) affected	0.000 sec
3	20:01:22	create table employee(employee_id int, employee_name varchar(50), gender enum("M","F"), age int, hire_date date, designation varchar(100), department_id int, location_id int, salary decimal (10.2))	0 row(s) affected	0.047 sec
4	20:01:22	create table department(department_ID int, department_Name varchar(100))	0 row(s) affected	0.046 sec
5	20:01:22	create TABLE location (location_ID int, location varchar(300))	0 row(s) affected	0.032 sec

The status bar at the bottom indicates 'Query Completed'.

2 Table alteration

Add new column

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'employees' selected. The main query editor contains the following SQL script:

```
13
14
15 • create table department(department_ID int,
16   department_Name varchar(100)
17
18 );
19 • create TABLE location (location_ID int,
20   location varchar(300)
21
22 );
23 • alter table employee
24   add column email varchar(50);
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	21:00:34	alter table employee add column email varchar(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec

The status bar at the bottom indicates 'Query Completed'.

Modify

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'employees' expanded, showing 'department', 'employee', and 'location' tables. The 'employee' table is selected, and its columns are listed: employees_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary, and email. The main editor shows a SQL query with the following statements:

```
15 • create table department(department_ID int,  
16 department_Name varchar(100)  
17 );  
18 •  
19 • create TABLE location (location_ID int,  
20 location varchar(300)  
21 );  
22 •  
23 • alter table employee  
24 add column email varchar(50);  
25 • ALTER TABLE Employee  
26 MODIFY COLUMN designation VARCHAR(50);
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	21:00:34	alter table employee add column email varchar(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec
2	21:03:00	ALTER TABLE Employees MODIFY COLUMN designation VARCHAR(50)	Error Code: 1146. Table 'employees.employees' doesn't exist	0.000 sec
3	21:03:46	ALTER TABLE Employee MODIFY COLUMN designation VARCHAR(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec

Query Completed

Drop

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'employees' expanded, showing 'department', 'employee', and 'location' tables. The 'employee' table is selected, and its columns are listed: employees_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary, and email. The main editor shows a SQL query with the following statements:

```
17 •  
18 •  
19 • create TABLE location (location_ID int,  
20 location varchar(300)  
21 );  
22 •  
23 • alter table employee  
24 add column email varchar(50);  
25 • ALTER TABLE Employee  
26 MODIFY COLUMN designation VARCHAR(50);  
27 • alter table employee  
28 drop column age;
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	21:00:34	alter table employee add column email varchar(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec
2	21:03:00	ALTER TABLE Employees MODIFY COLUMN designation VARCHAR(50)	Error Code: 1146. Table 'employees.employees' doesn't exist	0.000 sec
3	21:03:46	ALTER TABLE Employee MODIFY COLUMN designation VARCHAR(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
4	21:08:10	alter table employee drop column age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.079 sec

Query Completed

3 Table Renaming (RENAME):

Rename the "Departments" table to "Departments_Info".

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'employees' database selected. The 'departments_info' table is highlighted. The main editor shows a SQL query with the following statements:

```
20 location varchar(300)
21
22 });
23
24 alter table employee
25 add column email varchar(50);
26
27 ALTER TABLE Employee
28 MODIFY COLUMN designation VARCHAR(50);
29
30 alter table employee
31 drop column age;
32
33 RENAME TABLE
34 Department TO Departments_Info,
35 Location TO Locations;
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	21:00:34	alter table employee add column email varchar(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec
2	21:03:00	ALTER TABLE Employees MODIFY COLUMN designation VARCHAR(50)	Error Code: 1146. Table 'employees.employees' doesn't exist	0.000 sec
3	21:03:46	ALTER TABLE Employee MODIFY COLUMN designation VARCHAR(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
4	21:08:10	alter table employee drop column age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.079 sec
5	21:11:50	RENAME TABLE Department TO Departments_Info, Location TO Locations	0 row(s) affected	0.031 sec
6	21:12:09	SELECT * FROM employees.departments_info LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'employees' database selected. The 'locations' table is highlighted. The main editor shows a SQL query with the following statements:

```
20 location varchar(300)
21
22 });
23
24 alter table employee
25 add column email varchar(50);
26
27 ALTER TABLE Employee
28 MODIFY COLUMN designation VARCHAR(50);
29
30 alter table employee
31 drop column age;
32
33 RENAME TABLE
34 Department TO Departments_Info,
35 Location TO Locations;
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	21:00:34	alter table employee add column email varchar(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec
2	21:03:00	ALTER TABLE Employees MODIFY COLUMN designation VARCHAR(50)	Error Code: 1146. Table 'employees.employees' doesn't exist	0.000 sec
3	21:03:46	ALTER TABLE Employee MODIFY COLUMN designation VARCHAR(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
4	21:08:10	alter table employee drop column age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.079 sec
5	21:11:50	RENAME TABLE Department TO Departments_Info, Location TO Locations	0 row(s) affected	0.031 sec
6	21:12:09	SELECT * FROM employees.departments_info LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec

4 Truncate

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' panel with a tree view showing 'departments_info', 'employee', 'locations', 'Views', 'Stored Procedures', and 'Functions'. The 'employee' table is selected, and its columns are listed: employees_id (int), employee_name (varchar(50)), gender (enum('M','F')), hire_date (date), designation (varchar(50)), department_id (int), location_id (int), salary (decimal(10,2)), and email (varchar(50)).

The main query window contains the following SQL code:

```
1 create database employees;
2 use employees;
3 create table employee(employees_id int,
4 employee_name varchar(50),
5 gender enum("M","F"),
6 age int,
7 hire_date date,
8 designation varchar(100),
9 department_id int,
10 location_id int,
11 salary decimal (10.2)
12
13
```

The output window shows the execution results of the queries:

#	Time	Action	Message	Duration / Fetch
2	21:03:00	ALTER TABLE Employees MODIFY COLUMN designation VARCHAR(50)	Error Code: 1146. Table 'employees.employees' doesn't exist	0.000 sec
3	21:03:46	ALTER TABLE Employee MODIFY COLUMN designation VARCHAR(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
4	21:08:10	alter table employee drop column age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.079 sec
5	21:11:50	RENAME TABLE Department TO Departments_Info, Location TO Locations	0 row(s) affected	0.031 sec
6	21:12:09	SELECT * FROM employees.departments_info LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
7	21:15:43	truncate table employee	0 row(s) affected	0.031 sec

5 Database & Table Dropping (DROP): drop the table

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' panel with a tree view showing 'employees', 'departments_info', 'locations', 'Views', 'Stored Procedures', and 'Functions'. The 'employees' database is selected, and its tables are listed: departments_info, locations, Views, Stored Procedures, and Functions. The 'No object selected' message is displayed.

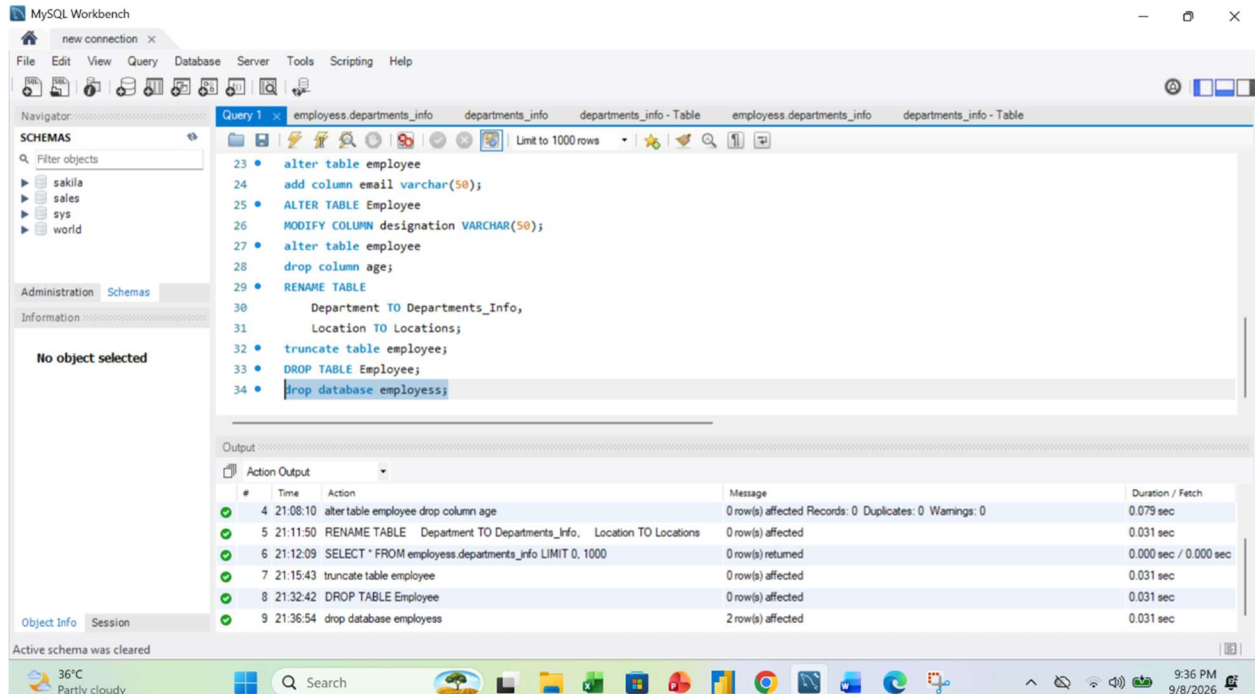
The main query window contains the following SQL code:

```
22 );
23
24 alter table employee
25 add column email varchar(50);
26
27 ALTER TABLE Employee
28 MODIFY COLUMN designation VARCHAR(50);
29
30 alter table employee
31 drop column age;
32
33 RENAME TABLE
34 Department TO Departments_Info,
35 Location TO Locations;
36
37 truncate table employee;
38
39 DROP TABLE Employee
```

The output window shows the execution results of the queries:

#	Time	Action	Message	Duration / Fetch
3	21:03:46	ALTER TABLE Employee MODIFY COLUMN designation VARCHAR(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
4	21:08:10	alter table employee drop column age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.079 sec
5	21:11:50	RENAME TABLE Department TO Departments_Info, Location TO Locations	0 row(s) affected	0.031 sec
6	21:12:09	SELECT * FROM employees.departments_info LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
7	21:15:43	truncate table employee	0 row(s) affected	0.031 sec
8	21:32:42	DROP TABLE Employee	0 row(s) affected	0.031 sec

Drop the data base



Constraint summary

Requirement	SQL constraint
Unique department ID	PRIMARY KEY
Department name cannot be null	NOT NULL
No duplicate department names	UNIQUE
Automatic location ID	AUTO_INCREMENT
Unique location	UNIQUE

Requirement	SQL constraint
Employee unique ID	PRIMARY KEY
Employee name mandatory	NOT NULL
Gender only M/F	CHECK (gender IN ('M','F'))
Age must be 18+	CHECK (age >= 18)
Default hire date	DEFAULT (CURRENT_DATE)
Department relationship	FOREIGN KEY
Location relationship	FOREIGN KEY